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Lesson 9: Clustering

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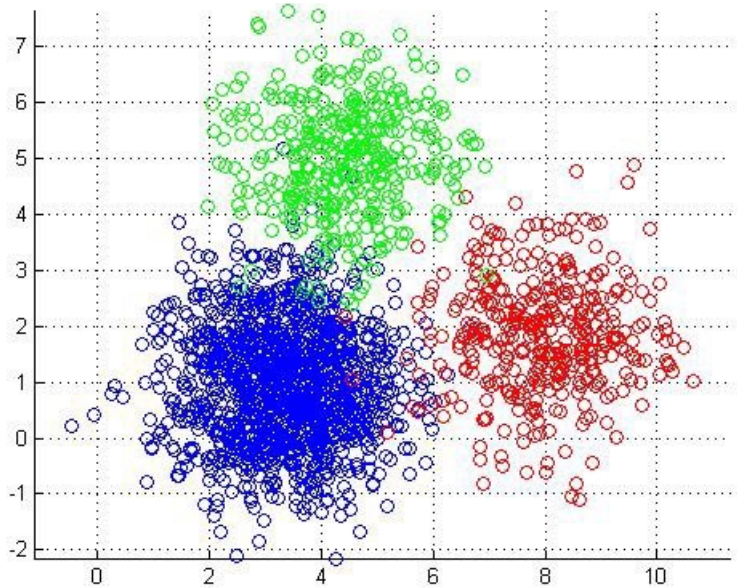
Clustering (聚类)

Unsupervised Learning: We do not have any labelled samples

Clustering: is the task of dividing the population or data points into a number of groups such that data points in the same groups are more similar to other data points in the same group than those in other groups. In simple words, the aim is to segregate groups with similar traits and assign them into cluster. （在无监督学习任务中研究最多，应用最广。既可以作为一个单独过程，也可以作为其他学习任务的前驱过程）

Difficulty: how to do performance measurement without ground truth, how to tune the parameters (hyperparameter)



Performance measurement

Clustering performance measurement, also known as clustering validity index

- **External index** （外部指标）

  Compare the clustering results with a reference model （例如将领域专家给出的划分作为参考模型）

  Such as Jaccard coefficient, FM index, Rand index

$$a = |SS|, \quad SS = \{(\boldsymbol{x}_i, \boldsymbol{x}_j) \mid \lambda_i = \lambda_j, \lambda_i^* = \lambda_j^*, i < j)\},$$

$$b = |SD|, \quad SD = \{(\boldsymbol{x}_i, \boldsymbol{x}_j) \mid \lambda_i = \lambda_j, \lambda_i^* \neq \lambda_j^*, i < j)\},$$

$$c = |DS|, \quad DS = \{(\boldsymbol{x}_i, \boldsymbol{x}_j) \mid \lambda_i \neq \lambda_j, \lambda_i^* = \lambda_j^*, i < j)\},$$

$$d = |DD|, \quad DD = \{(\boldsymbol{x}_i, \boldsymbol{x}_j) \mid \lambda_i \neq \lambda_j, \lambda_i^* \neq \lambda_j^*, i < j)\},$$

$$\mathrm{JC} = \frac{a}{a+b+c} \qquad \mathrm{FMI} = \sqrt{\frac{a}{a+b} \cdot \frac{a}{a+c}}. \qquad \mathrm{RI} = \frac{2(a+d)}{m(m-1)}$$

Performance measurement

☐ An example

样本	真实类别（λ^*）	聚类结果（λ）
x_1	A	1
x_2	A	1
x_3	B	2
x_4	A	2

样本对	$\lambda_i = \lambda_j$?	$\lambda_i^* = \lambda_j^*$?	属于哪一类
(x_1, x_2)	✅ (1=1)	✅ (A=A)	✅ a类
(x_1, x_3)	❌ (1≠2)	❌ (A≠B)	✅ d类
(x_1, x_4)	❌ (1≠2)	✅ (A=A)	✅ c类
(x_2, x_3)	❌ (1≠2)	❌ (A≠B)	✅ d类
(x_2, x_4)	❌ (1≠2)	✅ (A=A)	✅ c类
(x_3, x_4)	✅ (2=2)	❌ (B≠A)	✅ b类

Performance measurement

☐ An example

指标	样本对数量	样本对举例
a	1	(x_1, x_2)
b	1	(x_3, x_4)
c	2	$(x_1, x_4), (x_2, x_4)$
d	2	$(x_1, x_3), (x_2, x_3)$
总共	$C_4^2 = 6$ 对	

$$J = \frac{a}{a+b+c}$$

$$RI = \frac{a+d}{a+b+c+d}$$

$$FM = \sqrt{\frac{1}{2} \cdot \frac{1}{3}} = \sqrt{\frac{1}{6}} \approx 0.408$$

$$J = \frac{1}{1+1+2} = \frac{1}{4} = 0.25$$

$$RI = \frac{1+2}{1+1+2+2} = \frac{3}{6} = 0.5$$

聚一块的样本对中预测对的比例

查准率 & 查全率的综合表现

所有样本对中，聚类判断与真实一致的比例

Performance measurement

- **Internal index** （内部指标）

  Directly examine the clustering results without any reference model

  Such as DB index. Dunn index. etc.

$$\mathrm{avg}(C) = \frac{2}{|C|(|C|-1)} \sum_{1 \leqslant i < j \leqslant |C|} \mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j)$$ 平均距离

$$\mathrm{diam}(C) = \max_{1 \leqslant i < j \leqslant |C|} \mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) \ ,$$ 簇的直径

$$d_{\min}(C_i, C_j) = \min_{\boldsymbol{x}_i \in C_i, \boldsymbol{x}_j \in C_j} \mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) \ ,$$ 最小簇间距离

$$d_{\mathrm{cen}}(C_i, C_j) = \mathrm{dist}(\boldsymbol{\mu}_i, \boldsymbol{\mu}_j) \ ,$$

簇中心间距离

- DB 指数(Davies-Bouldin Index, 简称 DBI)

$$\mathrm{DBI} = \frac{1}{k} \sum_{i=1}^{k} \max_{j \neq i} \left(\frac{\mathrm{avg}(C_i) + \mathrm{avg}(C_j)}{d_{\mathrm{cen}}(\boldsymbol{\mu}_i, \boldsymbol{\mu}_j)} \right) .$$

- Dunn 指数(Dunn Index, 简称 DI)

$$\mathrm{DI} = \min_{1 \leqslant i \leqslant k} \left\{ \min_{j \neq i} \left(\frac{d_{\min}(C_i, C_j)}{\max_{1 \leqslant l \leqslant k} \mathrm{diam}(C_l)} \right) \right\} .$$

Basic idea:

- High intra-cluster similarity（簇内相似度)
- Low inter-cluster similarity（簇间相似度）

Distance calculation

The basic properties of distance metric:

非负性: $\mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) \geqslant 0$;
同一性: $\mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) = 0$ 当且仅当 $\boldsymbol{x}_i = \boldsymbol{x}_j$;
对称性: $\mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \mathrm{dist}(\boldsymbol{x}_j, \boldsymbol{x}_i)$;
直递性: $\mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_j) \leqslant \mathrm{dist}(\boldsymbol{x}_i, \boldsymbol{x}_k) + \mathrm{dist}(\boldsymbol{x}_k, \boldsymbol{x}_j)$

Common distance forms:
Minkowski distance （有序属性）

$$\mathrm{dist}_{\mathrm{mk}}(\boldsymbol{x}_i, \boldsymbol{x}_j) = \left(\sum_{u=1}^{n} |x_{iu} - x_{ju}|^p \right)^{\frac{1}{p}}$$

p = 2: Euclidean distance
p = 1: Manhattan distance

Distance calculation (continued)

- For **non-ordinal attributes (无序属性)**, **VDM** (Value Difference Metric) can be used

Let $m_{u,a}$ denote the number of samples whose value is a on attribute u, $m_{u,a,i}$ denote the number of samples whose value is a on attribute u in the i-th sample cluster, and k be the number of sample clusters, then two discrete values a and The VDM distance between b is

$$VDM_p(a,b) = \sum_{i=1}^{k} \left| \frac{m_{u,a,i}}{m_{u,a}} - \frac{m_{u,b,i}}{m_{u,b}} \right|^p$$

VDM 的核心思想是：比较两个离散值在各类别中的条件概率分布差异。

符号	含义
$m_{u,a}$	属性 u 取值为 a 的样本总数
$m_{u,a,i}$	属性 u 取值为 a，且类别为 i 的样本数量
k	样本的类别总数
p	幂指数，常取 1 或 2

Distance calculation (continued)

假设有属性“颜色”，其可能取值为 {红，蓝，黄}，用于预测“是否地震”（是 / 否），即二分类问题。

颜色	类别（是否地震）
红	是
红	是
红	否
蓝	否
蓝	否
黄	是
黄	否

红：3个样本（2是，1否）→ $P(\text{是}|\text{红}) = 2/3, P(\text{否}|\text{红}) = 1/3$

蓝：2个样本（0是，2否）→ $P(\text{是}|\text{蓝}) = 0, P(\text{否}|\text{蓝}) = 1$

$$VDM_1(\text{红},\text{蓝}) = |2/3 - 0| + |1/3 - 1| = 2/3 + 2/3 = \frac{4}{3}$$

- For mixed attributes, MinkovDM can be used

$$\mathrm{MinkovDM}_p(\boldsymbol{x}_i, \boldsymbol{x}_j) = \left(\sum_{u=1}^{n_c} |x_{iu} - x_{ju}|^p + \sum_{u=n_c+1}^{n} \mathrm{VDM}_p(x_{iu}, x_{ju}) \right)^{\frac{1}{p}}$$

Please remember

The goodness of clustering depends on
the opinion of the user （聚类好坏不存在绝对标准)

Common clustering methods

- **Prototype clustering**(Prototype-based clustering，原型聚类)
  - Hypothesis: The cluster structure can be described by a set of prototypes
  - Process: Initialize the prototype first, and then iteratively update the prototype
  - Representative: k-means clustering, learning vector quantization (LVQ), Gaussian mixture clustering
- **Density clustering**(Density-based clustering，密度聚类)
  - Hypothesis: The cluster structure can be determined by the tightness of the sample distribution
  - Process: From the perspective of sample density, examine the connectability between samples, and continuously expand clusters based on connectable samples
  - Representative: DBSCAN, OPTICS, DENCLUE
- **Hierarchical clustering** （层次聚类）
  - Hypothesis: can produce clustering results of different granularities
  - Process: divide the data set at different levels to form a tree-shaped cluster structure
  - Representative: AGNES (bottom-up), DIANA (top-down)

K-means

Basic concept:

***K*-means clustering** aims to partition *n* observations into *k* clusters in which each observation belongs to the cluster with the nearest mean (cluster centers or cluster centroid). Please not to be confused with K-nearest neighbors algorithm (KNN).

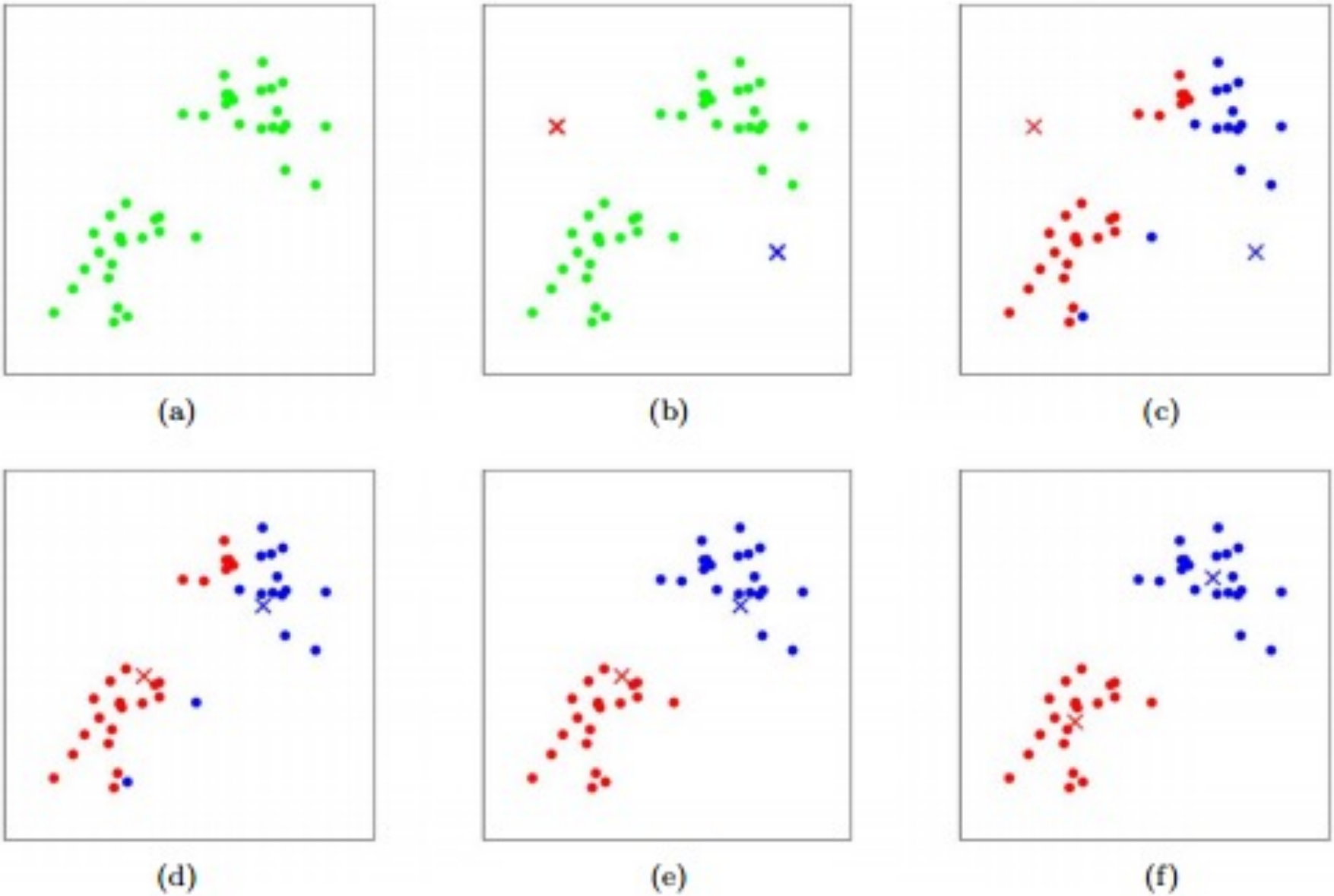
Centroid: In mathematics and physics, the **centroid** or **geometric center** of a plane figure is the **arithmetic mean** position of all the points in the figure.

Distance: Euclidean distance （欧氏距离）(perform normalization first)

Object： $min \sum_{\mathrm{i}=1}^{K} \sum_{x \epsilon Ci} dist(Ci, x)$^2

K-means

Illustration of K-means:



K-means

Advantages

- Simple, fast and suitable for regular datasets

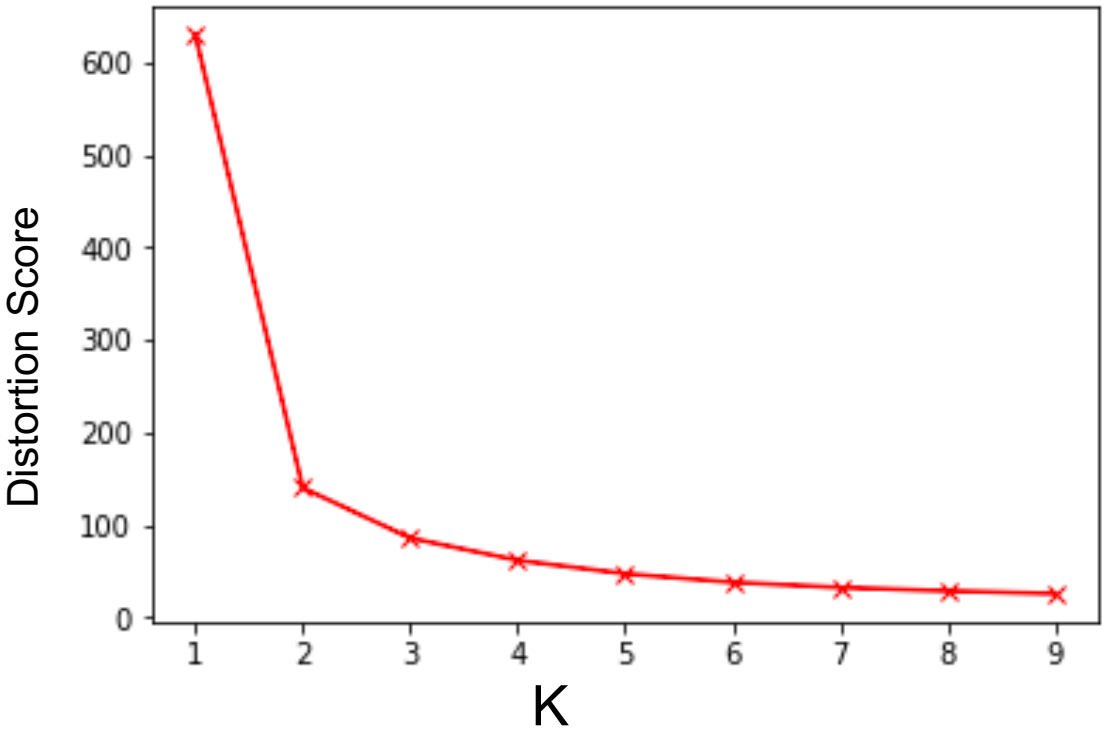
Disadvantages

- **K** value is very difficult to be determined (K值难确定)
- Complexity has a linear relationship with samples（复杂度与样本呈线性关系）O(n)
- Not easy to find any kind of clusters （很难发现任意形状的簇）

K-means

How to determine K?

Elbow method (clustering)



The distortion score: the sum of square distances from each point to its assigned center.

K-means

https://www.naftaliharris.com/blog/visualizing-k-means-clustering/

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Visualizing K-Means Clustering

January 19, 2014

Suppose you plotted the screen width and height of all the devices accessing this website. You'd probably find that the points form three clumps: one clump with small dimensions, (smartphones), one with moderate dimensions, (tablets), and one with large dimensions, (laptops and desktops). Getting an algorithm to recognize these clumps of points without help is called *clustering*. To gain insight into how common clustering techniques work (and don't work), I've been making some visualizations that illustrate three fundamentally different approaches. This post, the first in this series of three, covers the k-means algorithm. To begin, click an initialization strategy below:

How to pick the initial centroids?

选择不同的初始中心，会得到不同的聚类结果。

DBSCAN

DBSCAN stands for **d**ensity-**b**ased **s**patial **c**lustering of **a**pplications with **n**oise （具有噪声的基于密度的聚类方法）. It is able to find arbitrary shaped clusters and clusters with noise (i.e. outliers).

The main idea behind DBSCAN is that a point belongs to a cluster if it is close to many points from that cluster.

There are two key parameters of DBSCAN:

•**eps**: The distance that specifies the neighborhoods. Two points are considered to be neighbors if the distance between them are less than or equal to eps.

•**minPts:** Minimum number of data points to define a cluster.

•直接密度可达：若某点p在点q的r邻域内，且q是核心点则p-q直接密度可达

•密度可达：若有一个点的序列q_0、q_1、…q_k，对任意q_i-q_{i-1}是直接密度可达的，则称从q_0到q_k密度可达，这实际上是直接密度可达的“传播”

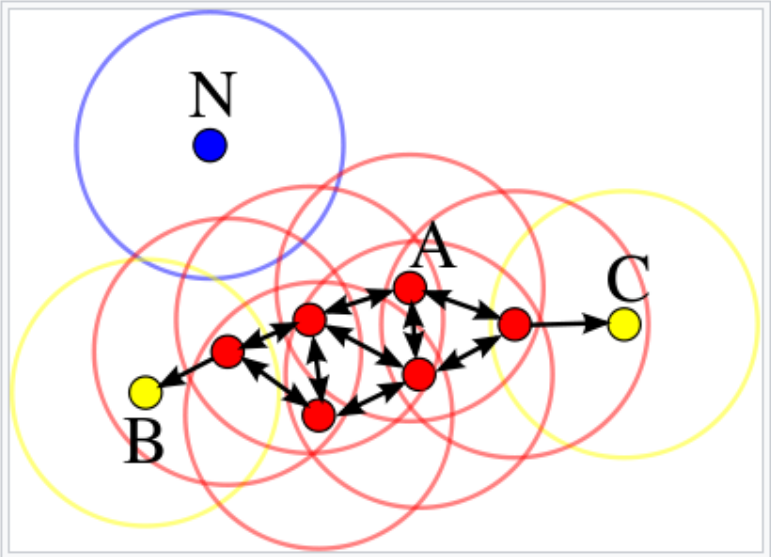
DBSCAN

Based on these two parameters, points are classified as core point, border point, or outlier:

•**Core point （核心点）:** A point is a core point if there are at least minPts number of points (including the point itself) in its surrounding area with radius eps.

•**Border point（边界点）:** A point is a border point if it is reachable from a core point and there are less than minPts number of points within its surrounding area.

•**Outlier （噪声点）:** A point is an outlier if it is not a core point and not reachable from any core points.



DBSCAN

工作流程：

参数D：输入数据集

参数eps：指定半径

可以根据K距离来设定：找到突变点
K距离：给定数据集P={p(i); i=0,1,..n},计算点P(i)到数据D的子集S中所有点之间的距离，按照从小到大顺序排序，d(k)就被称为k-距离

MinPts: 密度阈值

k-距离中的k值，一般取的小一些，多次尝试

输入：

D：一个包含 n 个对象的数据集

ε：半径参数

MinPts：领域密度阀值

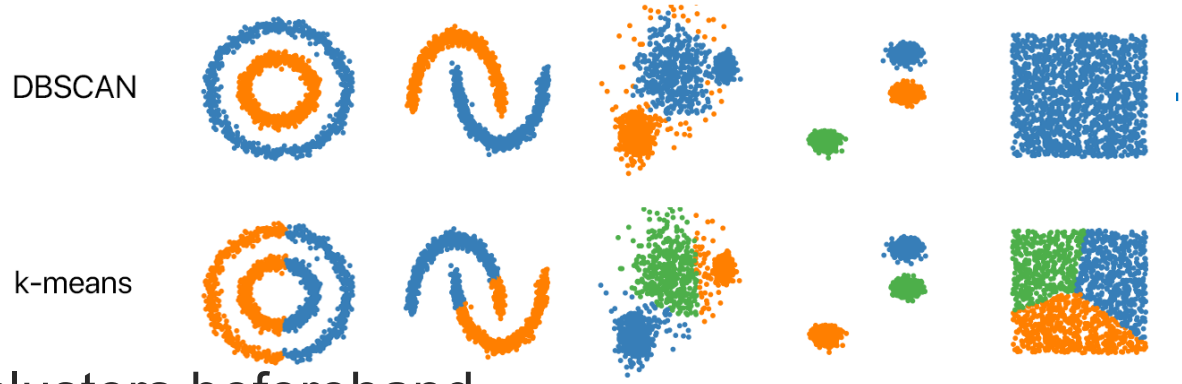
输出：基于密度的簇的集合

方法：

1. 标记所有对象为 unvisited;
2. Do
3. 随机选择一个 unvisited 对象 p；
4. 标记 p 为 visited；
5. If p 的 ε－领域至少有 MinPts 个对象
6. 创建一个新簇 C，并把 p 添加到 C；
7. 令 N 为 p 的 ε－领域 中的对象集合
8. For N 中每个点 p'
9. If p'是 unvisited;
10. 标记 p'为 visited;
11. If p'的 ε－领域至少有 MinPts 个对象，把这些对象添加到 N；
12. 如果 p'还不是任何簇的成员，把 p'添加到 C；
13. End for;
14. 输出 C；
15. Else 标记 p 为噪声；
16. Until 没有标记为 unvisited 的对象；

https://blog.csdn.net/qq_39293290

DBSCAN



Pros and Cons of DBSCAN

Pros:

•Does not require to specify number of clusters beforehand.
•Performs well with arbitrary shapes clusters.
•DBSCAN is robust to outliers and able to detect the outliers.

Cons:

•In some cases, determining an appropriate distance of neighborhood (eps) is not easy and it requires domain knowledge.

•If clusters are very different in terms of in-cluster densities, DBSCAN is not well suited to define clusters. The characteristics of clusters are defined by the combination of eps-minPts parameters. Since we pass in one eps-minPts combination to the algorithm, it cannot generalize well to clusters with much different densities.

DBSCAN

http://www.naftaliharris.com/blog/visualizing-dbscan-clustering/

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Visualizing DBSCAN Clustering

January 24, 2015

A previous post covered clustering with the k-means algorithm. In this post, we consider a fundamentally different, density-based approach called DBSCAN. In contrast to k-means, which modeled clusters as sets of points near to their center, density-based approaches like DBSCAN model clusters as high-density clumps of points. To begin, choose a data set below:

What kind of data would you like?

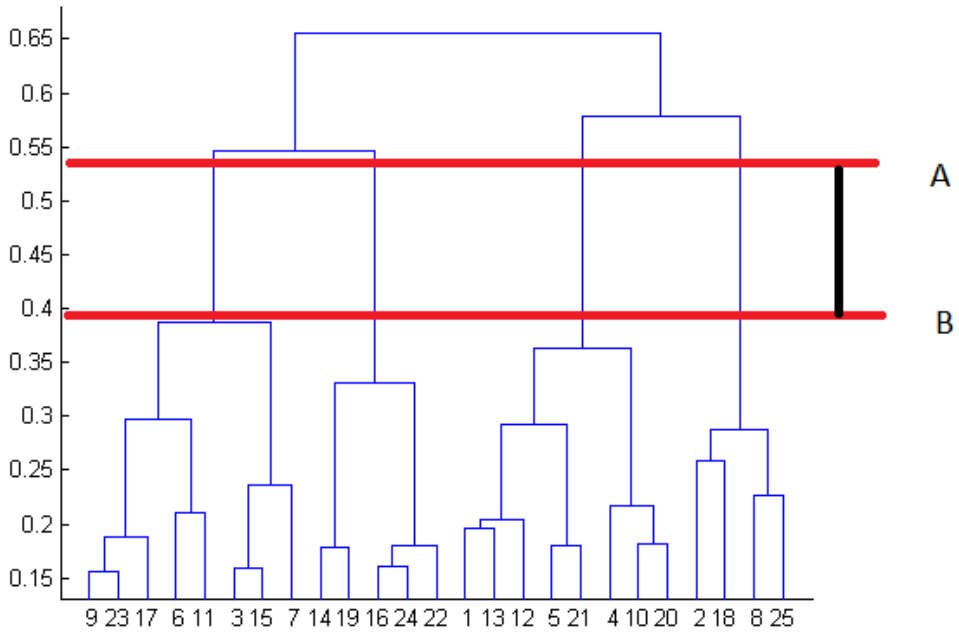
Uniform Points	Gaussian Mixture	Smiley Face
Density Bars	Packed Circles	Pimpled Smiley
DBSCAN Rings	Example A	

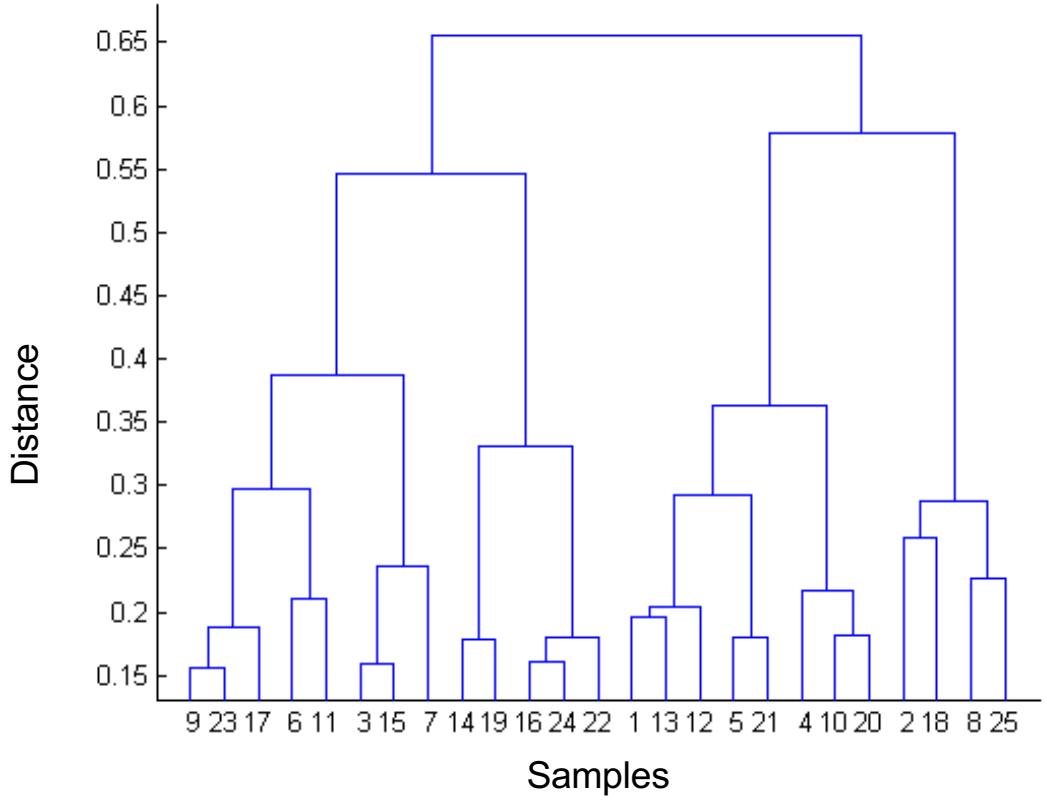
Hierarchical clustering （层次聚类）

Hierarchical clustering, as the name suggests is an algorithm that builds hierarchy of clusters. This algorithm starts with all the data points assigned to a cluster of their own. Then two nearest clusters are merged into the same cluster. In the end, this algorithm terminates when there is only a single cluster left.

不需要提前设定簇数 **K**，也不基于原型或密度，而是基于样本之间的距离逐步聚合。

The results of hierarchical clustering can be shown using dendrogram.
The dendrogram （树状图） can be interpreted as:





Hierarchical clustering

- Hierarchical clustering can't handle big data well but K Means clustering can. This is because the time complexity of K Means is linear i.e. O(n) while that of hierarchical clustering is quadratic i.e. O(n2).

- In K Means clustering, since we start with random choice of clusters, the results produced by running the algorithm multiple times might differ. While results are reproducible in Hierarchical clustering.

- K Means is found to work well when the shape of the clusters is hyper spherical (like circle in 2D, sphere in 3D).

- K Means clustering requires prior knowledge of K i.e. no. of clusters you want to divide your data into. But, you can stop at whatever number of clusters you find appropriate in hierarchical clustering by interpreting the dendrogram

Hierarchical clustering

Clustering has a large no. of applications spread across various domains. Some of the most popular applications of clustering are:

- Recommendation engines
- Market segmentation
- Social network analysis
- Search result grouping
- Medical imaging
- Image segmentation
- Anomaly detection

假如你在做一个音乐推荐系统：

•你将所有用户基于听歌风格做层次聚类；

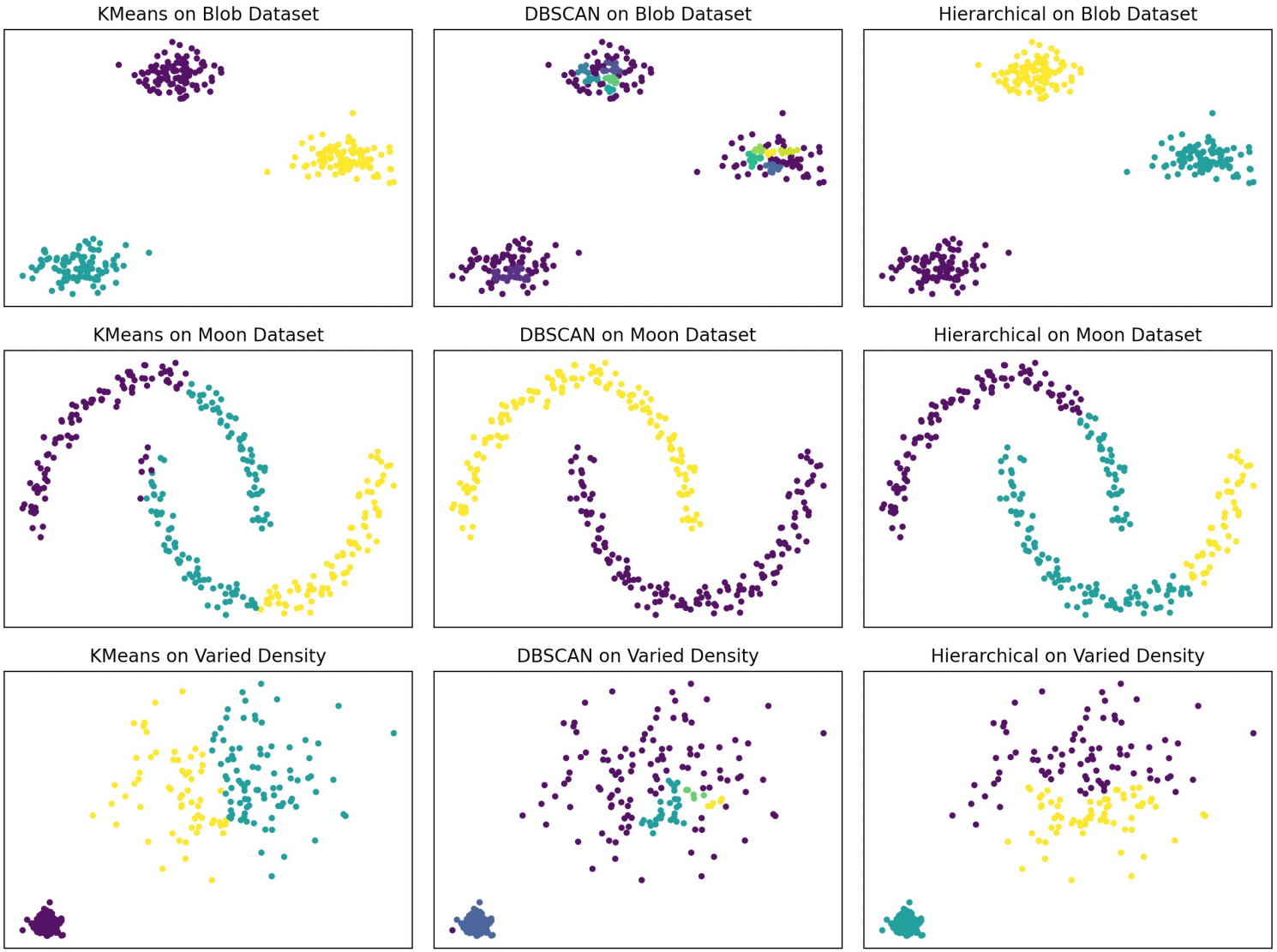
•发现有一簇用户偏好轻音乐，而其中又可以细分为古典轻音乐和新世纪轻音乐；

•针对新加入的用户，可以先判断他属于哪一大簇，再在对应子簇中做个性化推荐。

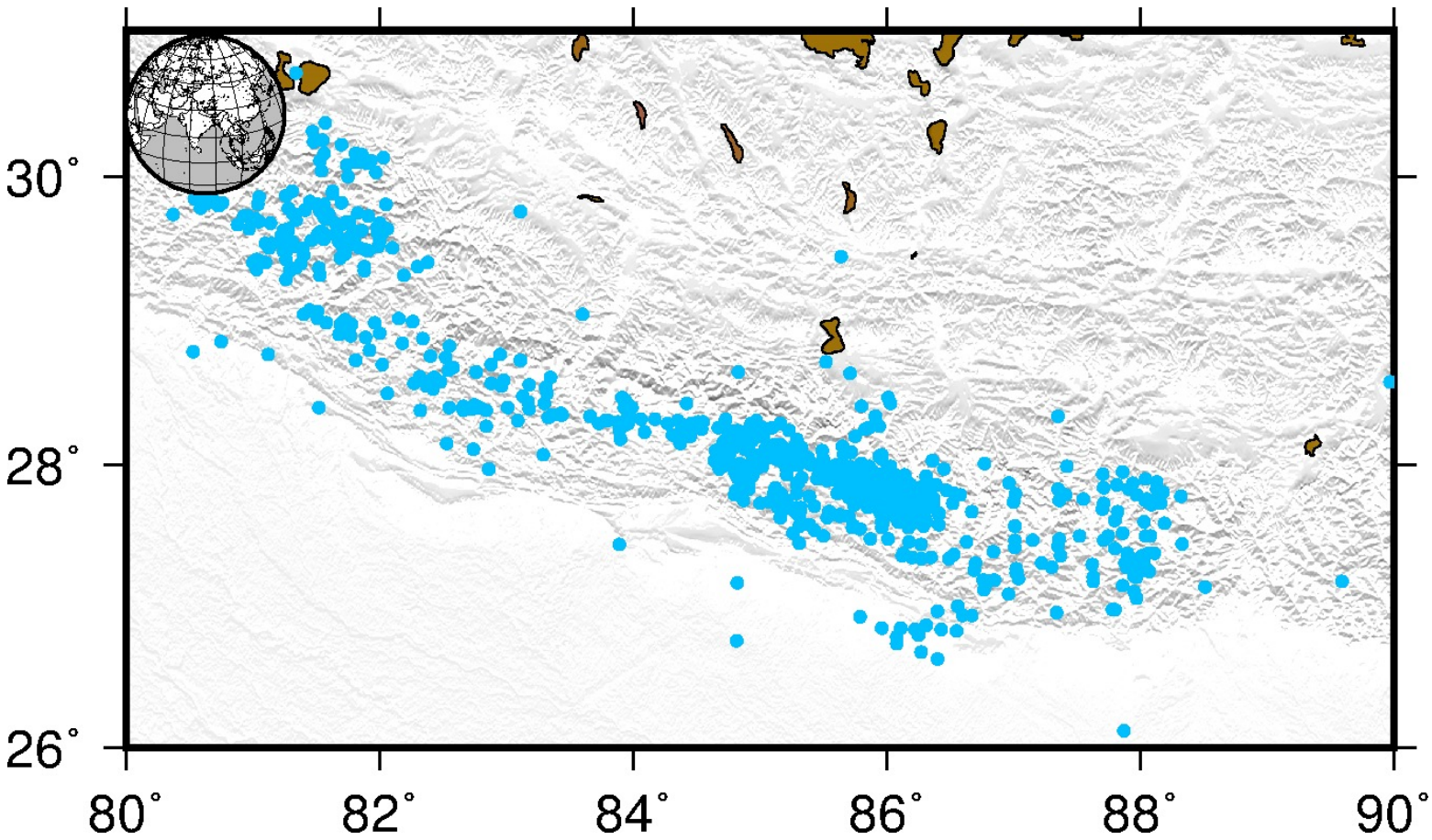
这就比一开始就用 **K-means** 设定 **5** 类，来得更灵活、更贴近用户行为的真实结构。

三种聚类方法对比总结

特征 / 算法	K-means	DBSCAN	Hierarchical Clustering
🔢 是否需预设聚类数	✅ 需要指定 K	❌ 不需要	❌ 不需要（可后期剪树选择簇数）
📏 聚类形状假设	类簇为**球形**（等方差）	任意形状都可以	任意形状，依赖距离度量
🌪️ 对异常值敏感性	❗ 敏感，异常值会扭曲质心	✅ 能识别并剔除异常值（噪声点）	❗ 对异常值敏感
🧮 算法复杂度	**O(n)**，线性，适合大数据	O(n log n) ~ O(n^2)，取决于索引结构	**O(n^2)**，不适合大数据
🌀 聚类稳定性	❌ 初始点随机，结果不稳定	✅ 非常稳定	✅ 稳定（构建单棵树）
🎯 适用场景	数据结构规则、聚类数已知	数据密度分布不均，有噪声	需解释聚类层级结构、小样本分析
📤 输出结构	一组聚类结果（平面结构）	一组聚类结果（含噪声）	一棵聚类树（dendrogram）
💻 参数	K 值	`eps`，`minPts`	距离度量方式（单链、全链、平均链等）



数据集类型	结构特征	适合的聚类方法
Blob	球形、均匀分布	KMeans
Moon	非线性结构	DBSCAN、层次聚类
Varied Density	密度差异大	层次聚类（部分）、DBSCAN



Earthquake clustering
by k-means